LOVELY PROFESSIONAL UNIVERSITY

Sample Question Set - Unit 6

Course Code: CSE276
Course Title: ARTIFICIAL INTELLIGENCE FOUNDATIONS
Course Outcome: CO6
Paper: Sample Subjective Questions
Maximum Marks: 10 x 10 = 100

Unit: 6
Unit Title: AI Development Ecosystem and Future Trends
Unit Mapped RBT: L3 (Apply)
Total Questions: 10
Suggested Duration: Not timed - practice set

Subjective Questions (10 x 10 = 100 Marks)

Q1. Apply the Python ecosystem to a small machine learning project on a table of records. (10 Marks)

(a) Name the libraries you would use at each stage, from loading the data to reporting the result.

(b) Explain why a deep learning framework is not needed for this project, and when it would be.

[L3 - Apply]

Q2. Compare TensorFlow and PyTorch as frameworks for building neural networks. (10 Marks)

(a) State what the two have in common and one respect in which they differ.

(b) Use a stated project requirement to select one of the two, and give the reason.

[L3 - Apply]

Q3. Apply a vector database to an assistant that must answer from a department's own documents. (10 Marks)

(a) State what is stored in the database and how a question is matched against it.

(b) Use the size of the collection to decide between an exact scan and an index, and justify the choice.

[L3 - Apply]

Q4. Examine what happens when the embedding model of a retrieval system is replaced after the index has been built. (10 Marks)

(a) State why the stored vectors and the new query vectors can no longer be compared.

(b) Describe the steps needed to bring the system back into a working state.

[L3 - Apply]

Q5. Use the Artificial Intelligence development workflow to plan a project from the first request to release. (10 Marks)

(a) Set out the stages in order, stating what each one produces for the stage that follows.

(b) Explain what is measured before release, and what a poor measurement would prompt you to do.

[L3 - Apply]

Q6. Compare exact search with approximate search inside a vector store. (10 Marks)

(a) State how each of the two finds the passages nearest to a question.

(b) Use the size of the store to choose between them, and state what the faster option gives up.

[L3 - Apply]

Q7. Demonstrate how Hugging Face, Ollama and a pipeline tool fit together in one grounded assistant. (10 Marks)

(a) State what each of the three contributes to the finished system.

(b) Use a department that may not send data outside the building to justify the arrangement you propose.

[L3 - Apply]

Q8. Examine what Artificial General Intelligence would mean, and how a claim to have reached it should be judged. (10 Marks)

(a) State what distinguishes general intelligence from the narrow systems in service today.

(b) Use a system that performs superbly on one benchmark to explain why that is not sufficient evidence.

[L3 - Apply]

Q9. Apply the idea of sustainable Artificial Intelligence to the design of a service that will run for several years. (10 Marks)

(a) State two decisions at build time and two at serving time that affect the energy consumed.

(b) Use a stated volume of queries to explain which of those decisions matters most.

[L3 - Apply]

Q10. Relate the reproducibility of a result to the practices a development team should adopt. (10 Marks)

(a) State four things that must be recorded if a result is to be reproduced months later.

(b) Use a project that cannot repeat last month's figure to show which of the four was most likely missing.

[L3 - Apply]

----- End of Sample Question Set -----